1. **Project Title**

WildTrack: Your Guide to Wildlife in National Parks

2. **Project Summary**

Our project is a relational database-centric web application designed to streamline wildlife-centric trip planning for national park enthusiasts. Currently, identifying where and when to see specific animals in national parks requires tedious research across multiple disconnected sources. Our platform centralizes this information, allowing users to search by species to find corresponding national parks and the precise coordinates where those animals have been previously sighted. Users can also browse comprehensive national park profiles to see their resident biodiversity. The system bridges the gap between static geographical data and dynamic ecological observations.

The core functionality includes a user authentication system where members can manage their sighting history. A key feature is the “Community Sightings Map,” which visualizes user-uploaded data alongside official park information. By allowing users to submit forms with species, time, and location data, the application generates a real-time pin-map of wildlife activity. This project demonstrates complex relational modeling between users, national parks, and species, providing a comprehensive tool for both casual tourists and serious wildlife enthusiasts.

Detailed Functionality:

1. User Authentication & Personal Dashboard
   A secure login system allows members to manage their profile and maintain a personalized sighting journal
2. Multi-Dimensional Search
   A comprehensive search bar enables users to query by species name or park name.
3. Interactive Sighting Map
   Visualization of common seen locations. This map overlays official park data with real-time user-uploaded sightings, pinpointing hotspots for wildlife activity
4. Crowdsourced Data Contribution
   Authenticated users can contribute to the community by submitting a sighting form, including species, time and GPS coordinates.
5. Global Directories
   The platform features dedicated exploration pages, including a full index of all National Parks and a comprehensive species catalog, ensuring users to browse the entire dataset.

3. What would be a good creative component that can improve the functionality of your application? To get a better sense of what a creative component is, these are **technically challenging features** that improve the user experience of your application. Some examples include interactive visualization (using several packages with some level or engineering), using several APIs to support some information presentation or using smart transformations to process data. Some examples that are NOT creative components include: software features that are completed with a few lines of code (e.g. adding a google maps iframe). Again, if you are unsure, discuss with your project TA.

We are integrating and transforming real species record data downloaded from iNaturalist to our database project to support an interactive species and mapping experience.

Our application allows users to filter wildlife observations and visualize the results on an interactive map. When no filters are applied, the interactive map will provide a default view that displays all available species observations across national parks. When a user selects a specific species and a national park, the system queries the underlying dataset and displays the exact observation coordinates as pins on the map, representing real locations where that species has been observed and showing descriptions written by other users.

This requires backend data processing to clean and organize large-scale observation records, and efficient querying and transformation of location data for visualization. The map updates in real time based on user selections, making this a technically challenging feature that combines data transformation, filtering logic, and interactive visualization.

4. **Usefulness**. Explain as clearly as possible why your chosen application is useful. What are the basic functions of your web application? (What can users of this website do? Which simple and complex features are there?). Make sure to answer the following questions: Are there any similar websites/applications out there? If so, what are they, and how is yours different?

This web application is designed for travelers and wildlife lovers who want to plan trips around specific animals, not just destinations. Most existing travel platforms require users to choose a national park first and then discover what wildlife might be there. Our application reverses this workflow by letting users start with their wildlife interest and immediately discover where they can see it. Users can search for animals such as

“Brown Bear” or “Bald Eagle” and instantly view all national parks where that species is commonly found, along with an interactive map showing real sighting locations. This saves the need to cross reference multiple websites, blogs or forums to plan a wildlife trip planning faster and more intuitively.

The basic features include user authentication, a searchable and sortable database of animals and national parks, and detailed pages for each species and park. Users can browse all animals or all national parks in the dataset, view cleanly organized detail pages, and explore common sighting locations on a map. After logging in, users can submit their own animal sightings through a form, which will be saved to their personal account and visualized as pins on the map.

While similar platforms exist, our application is still unique. The official National Park Service website provides park information but lacks species-first discovery and community driven sighting data. Wildlife platforms like iNaturalist focus on biological documentation rather than travel planning and park comparison. But our application uniquely combines park exploration, wildlife discovery, and user generated sightings into a clear visualized experience for planning trips.

5. **Realness**. We want you to build a real application. So, make sure to locate real datasets. Describe your data sources (Where is the data from? In what format [csv, xls, txt,...], data size [cardinality and degree], what information does the data source capture?). It would be hard to satisfy stage 2 requirements with one dataset. Thus, **we strongly recommend identifying at least two different data sources for your project, part of the dataset (i.e., one of the table) should have at least 1K data.**

National Parks
National Park Species
I naturalist real species location data

The National Parks dataset is published by the National Park Service and provides official geographic information and boundaries for U.S. national parks. The dataset is available in multiple formats, including CSV, GeoJSON, and contains approximately 437 records. Key attributes include park identifiers, park names, and spatial boundary geometries. In our application, we use the GeoJSON format to represent national park boundaries for map visualization, while the CSV format is used to store and manage park metadata such as park names and identifiers in the backend database.

The National Park Species Dataset (2024) published by the National Park Service. This dataset records verified animal and plant species present in individual national parks, based on observations, vouchers, and reports. It is provided in CSV and contains 40000+ rows. Each record includes detailed taxonomic information (scientific name, common name, family, order), park identifiers, species occurrence status, abundance, nativeness, seasonality, and conservation classifications.

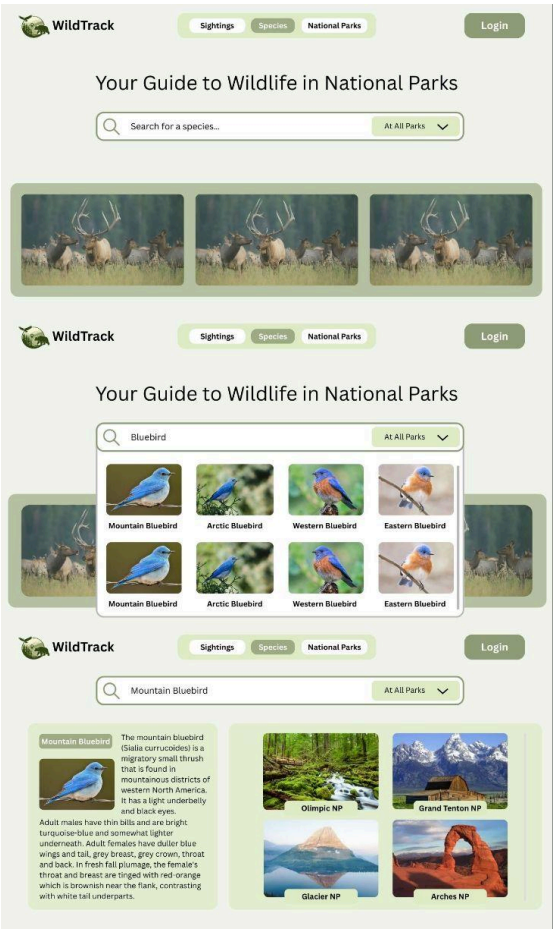
The iNaturalist dataset provides species observation records, including precise geographic coordinates where species have been observed. Each record captures information such as species identification, latitude and longitude, observation timestamps, and observer metadata, allowing us to visualize actual species sighting locations as map pins and support species-level filtering within national parks.

6. A detailed description of the **functionality** that your website offers. This is where you talk about what the website delivers. Talk about how a user would interact with the application (i.e., things that one could create, delete, update, or search for). **Read the requirements for stage 4 to see what other functionalities you want to provide to the users.** You should include:

1. **A clear list of the functionality:** Who [add/insert/update/delete] what types of data in your application when? (e.g., the user would submit a form with [what data] when they [want to do something].)
   i. User authentication:When user sign up, add username and password to users table (Create) and when user login, check users to verify. (Read)
   ii. Search bar: The application provides a search bar that allows users to search species by name and filter results by species category and national park. (Read)
      1. The results are displayed as a list of species, and selecting a species shows its introduction and all national parks in which the species has been present.
      2. When a user selects a national park, the application displays information about the park along with an interactive map showing the specific locations of recorded sightings within the park. Also, a visualization that summarizes sighting patterns by time of day and by month, based on aggregated user-submitted data will be displayed.
   iii. Wildlife Sighting Log: When users encounter a plant, animal, or insect, they submit a sighting through a form. (Create)
      1. Sightings are stored in a table with species name, date/time, weather, location, and optionally a photo. Over time, this builds a collaborative species discovery map, which can be explored by users.
      2. Users can update or delete their sightings (Update & Delete)

iv. User page: Users are able to see the list of species they recorded in their account. (Read)
v. Species page
vi. National parks page

2. **A low-fidelity UI mockup**: What do you imagine your final application's interface might look like? A PowerPoint slide or a pencil sketch on a piece of paper works!



3. **Project work distribution**: Who will be responsible for each of the tasks or subtasks?
   Explain how backend systems will be distributed across members. Be as specific as possible as this could be part of the final peer evaluation metrics.

Yueyao: Manages animal and park data features. Builds listing, search, filtering, and detail pages.

Leyi: Handles user system and data flow. Builds authentication, user dashboards, and manages user-submitted records from database to UI. Supports documentation and final deployment.

Meixian: Owns sighting and map features. Implements location-based data, map visualization, and related backend and frontend logic.

Junyi: Oversees system architecture, database design, dataset integration, and backend, frontend integration across all stages.